Step 0: Initial Document Check

- **0.1:** The document is an RCC structural drawing of "FOUNDATIONS" only.
- **0.2:** Site location: MARKAZ NAGARA.
- **0.3:** All compliance checks will be based only on IS 456:2000 and SP 34.

Step 1: Locate the "NOTES" Section

- **Status:** Compliant (A "NOTE" section is present, though some information is in a table labeled "CLEAR COVER" and "LAP LENGTH" and other general notes.)

Step 2 & 3: Extract and Verify Design Parameters from "NOTES"

Criteria	Extracted Value	Compliance Check	Status
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1. Grade of Concrete	M25	M25 is a common grade for foundations. Suitability for environmental exposure cannot be fully verified without specific site exposure conditions, but M25 is generally adequate for moderate conditions.	Cannot Verify
2. Reinforcement Bars	STEEL-FY500 (Fe 500). Stirrups are mentioned for columns (e.g., 8MM Ø 4 LEGGED STIRRUPS AT 6" C/C).	Fe 500 is a standard and compliant grade. Shear reinforcement (stirrups) is mentioned.	Compliant
3. Lap Length	8mm: 400mm, 10mm: 500mm, 12mm: 600mm, 16mm: 800mm, 20mm: 1000mm, 25mm: 1250mm. These values correspond to 50 times bar diameter (50d).	The lap lengths are specified as 50d, which is a common and generally compliant practice.	Compliant
4. Clear Cover	Footing: 50mm, Column: 38mm, Beam: 25mm, Slab: 15mm.	Footing cover of 50mm is the minimum as per IS 456:2000. If PCC is not specified, a higher cover (70-75mm) is preferred. Column cover of 38mm is below the typical 40mm minimum for columns. PCC is mentioned in the column footing section (P.C.C. IN C.C. 1:4:8).	Non-Compliant (Column cover)
5. Development Length (Ld)	Not explicitly stated as "Development Length". However, lap length is given as 50d, which is often related to development length.	Development length is not explicitly mentioned. While lap length is 50d, Ld should be specified separately or derived.	Missing Information
6. Safe Bearing Capacity (SBC) of soil	230 KN/SQM	SBC is mentioned. PCC is mentioned in the column footing section.	Compliant
7. Seismic Zone and Wind Load	Not mentioned.	This information is mandatory for structural design.	Missing Information
8. Building Limitations	"GROUND FLOOR + 2 FLOORS = 3 SLABS + ROOF TRUSS"	The number of floors is specified.	Compliant

9. Structure's Purpose	RESIDENTIAL BUILDING	The purpose is mentioned.	Compliant
10. Floor Heights	Not explicitly mentioned, but the section shows "GROUND LEVEL", "FIRST FLOOR", "SECOND FLOOR", "TERRACE", "ROOF TRUSS". Specific heights between floors are not given.	Floor heights are not explicitly stated.	Missing Information
11. Schedule of Footings	A "FOOTING TYPE" table is present, listing F1, F2, F3, F4 with bed size, footing size, depth, and reinforcement.	The drawing is consistent with the "FOOTING TYPE" table.	Compliant
12. Footing Type	Isolated footings (F1, F2, F3, F4).	Isolated footings are used for a residential building (low-to-medium rise).	Compliant
13. Reinforcement in High-Rise Buildings	The building is described as "GROUND FLOOR + 2 FLOORS", which is not considered high-rise.	Not Applicable	Not Applicable
14. Raft Foundation Reinforcement	Raft foundation is not used.	Not Applicable	Not Applicable
15. Lift Design	No lift is indicated in the drawing.	Not Applicable	Not Applicable
16. Soil Improvement	"5'0' OR HARD STRATA" is mentioned in the column footing section, implying natural hard strata rather than specific improvement.	No explicit soil improvement methods are detailed.	Not Applicable
17. Column Ties	Column ties (stirrups) are shown in the column details (e.g., "RINGS: 8MM Ø 4 LEGGED STIRRUPS AT 6" C/C"). They appear continuous in the schematic.	Column ties are specified and appear continuous.	Compliant
18. Plan of Ties	A separate plan for ties is not explicitly provided, but tie configurations are shown within the column details.	A separate plan for ties is not present.	Missing Information
19. Outer Ties Check	Column details specify tie diameters and spacing (e.g., 8mm dia stirrups at 6" c/c). Percentage of steel in columns is not explicitly stated in the notes but can be calculated from the bar schedule. Footing reinforcement is given (e.g., 12MM DIA BARS AT 6" C/C BOTH WAYS).	Percentage of steel in columns and footings is not explicitly stated in the notes. Column steel percentage needs to be calculated to verify compliance (>=0.8%). Footing reinforcement is provided.	Missing Information
21. Steel Curtailment	The building is low-rise (G+2), so curtailment might not be as critical or explicitly detailed as in high-rise structures. No specific details on curtailment are provided.	Not Applicable	Not Applicable

Step 5: Report Missing or Wrong Information

Grade of Concrete: Cannot fully verify suitability for environmental exposure without specific site conditions.

Clear Cover: Column cover of 38mm is below the typical 40mm minimum as per IS 456:2000.

Development Length (Ld): Not explicitly mentioned.

Seismic Zone and Wind Load: Mandatory information is missing.

Floor Heights: Specific heights between floors are not provided.

Plan of Ties: A separate plan for ties is not present.

Outer Ties Check: Percentage of steel in columns and footings is not explicitly stated in the notes.

Cross-Section Area: Not specified whether gross or effective cross-section area is used.

Maximum Steel Percentage in Columns: Not explicitly stated in the notes.

Summary of Compliance

- **Total Criteria Evaluated:** 22
- **Compliant Items:** 7
- **Non-Compliant Items:** 1
- **Missing Information Items:** 7
- **Cannot Verify:** 1
- **Not Applicable:** 6
- **Overall Verdict:** The drawing has several missing pieces of information and one non-compliant item (column clear cover) that need to be addressed for full compliance with IS 456:2000 and SP 34.